

Python Refresher - 1

- First Python Program
 - Hierarchy of programming languages (Bottom to top)
 - Everything in Python that has a value is an Object
 - Arithmetic Operations
 - Identifier
 - Mutable Immutable
 - Iteration Protocol
 - Functions
-

HOW'S THE
JOSH?

Python does not have ~~data types~~ primitive

↓
F C C s
↓ ↓ ↓
Foot class citizens.

Byte code

Rules :-

- ① class is going to start at 9:02 PM.
- ② we will break for a 5 mins after 1 hr (10:00)
- ③ we will finish class by 11:00 PM, from 11:00 to 11:30, we will having DCS.
- ④ I am going to repeat most of concepts twice, and some complex concepts even thrice.
- ⑤ Assignments + Feedback (1-5)

IDE :-

Pycharm
VScode
Anaconda

Jupyterhub

Google Colab

→ Online

① Python is a (dynamically typed language)

↓
don't need to pre define
the data types before hand.

$$\textcircled{1} \quad 1 + 2 = 3$$
$$\textcircled{11} \quad 1 + 2 \textcircled{11} = 1 + 2$$

② Python is an interpreted language.

Compiled

① Translates the entire
source code into
MLC code

② Faster after
compilation

③ Ex: C, C++,
Java

④ All errors are
shown after compilation

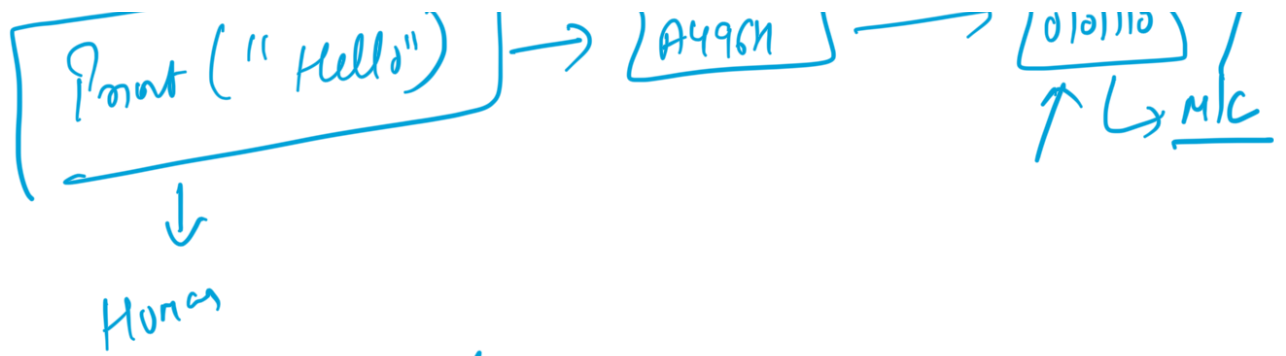
Interpreted

① Translates and
executes the code line
by line.

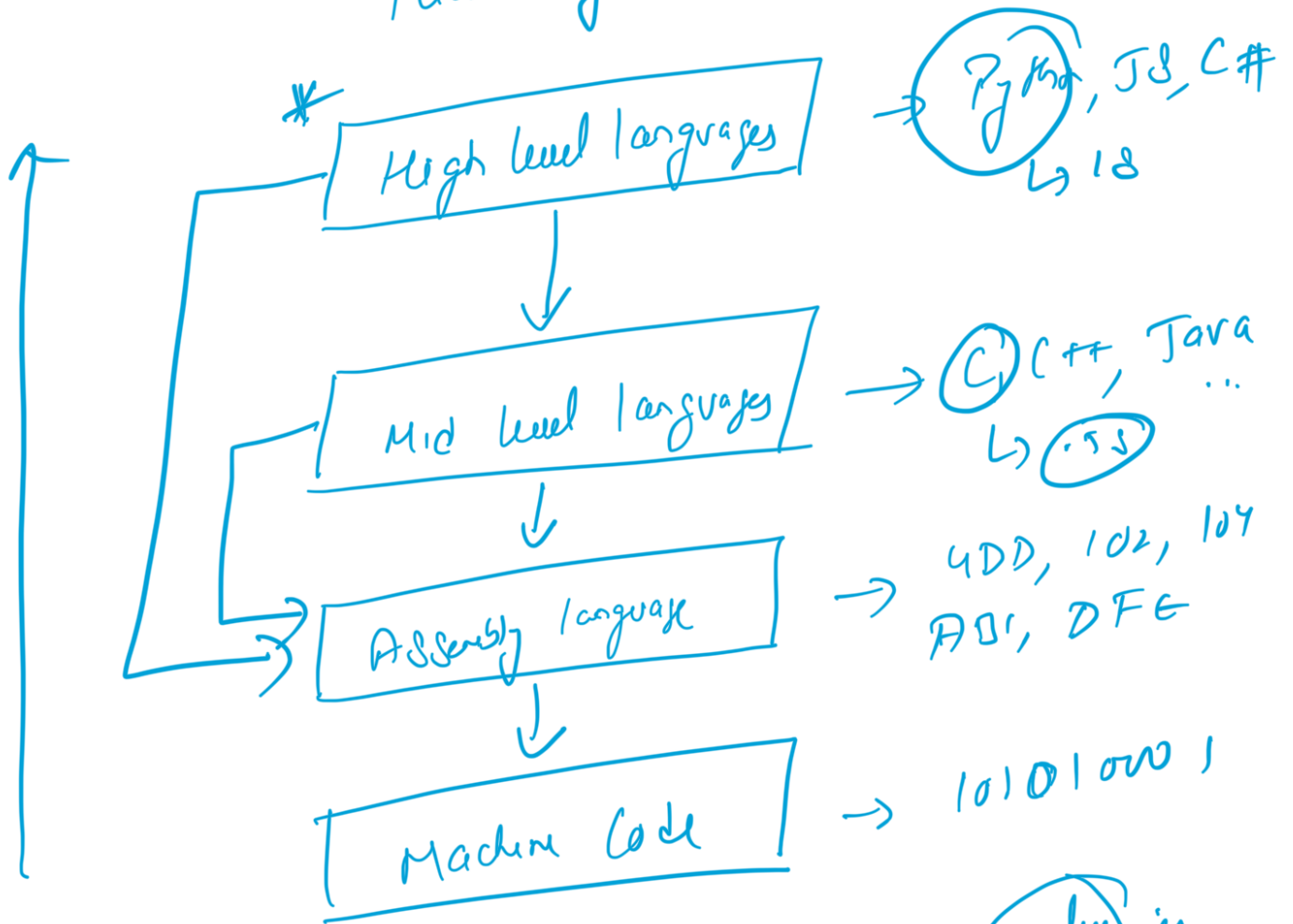
② Slower, as each line
is translated into execution

③ Ex: Python, Ruby,
JavaScript, PHP

④ Errors are shown
immediately when line
is executed.

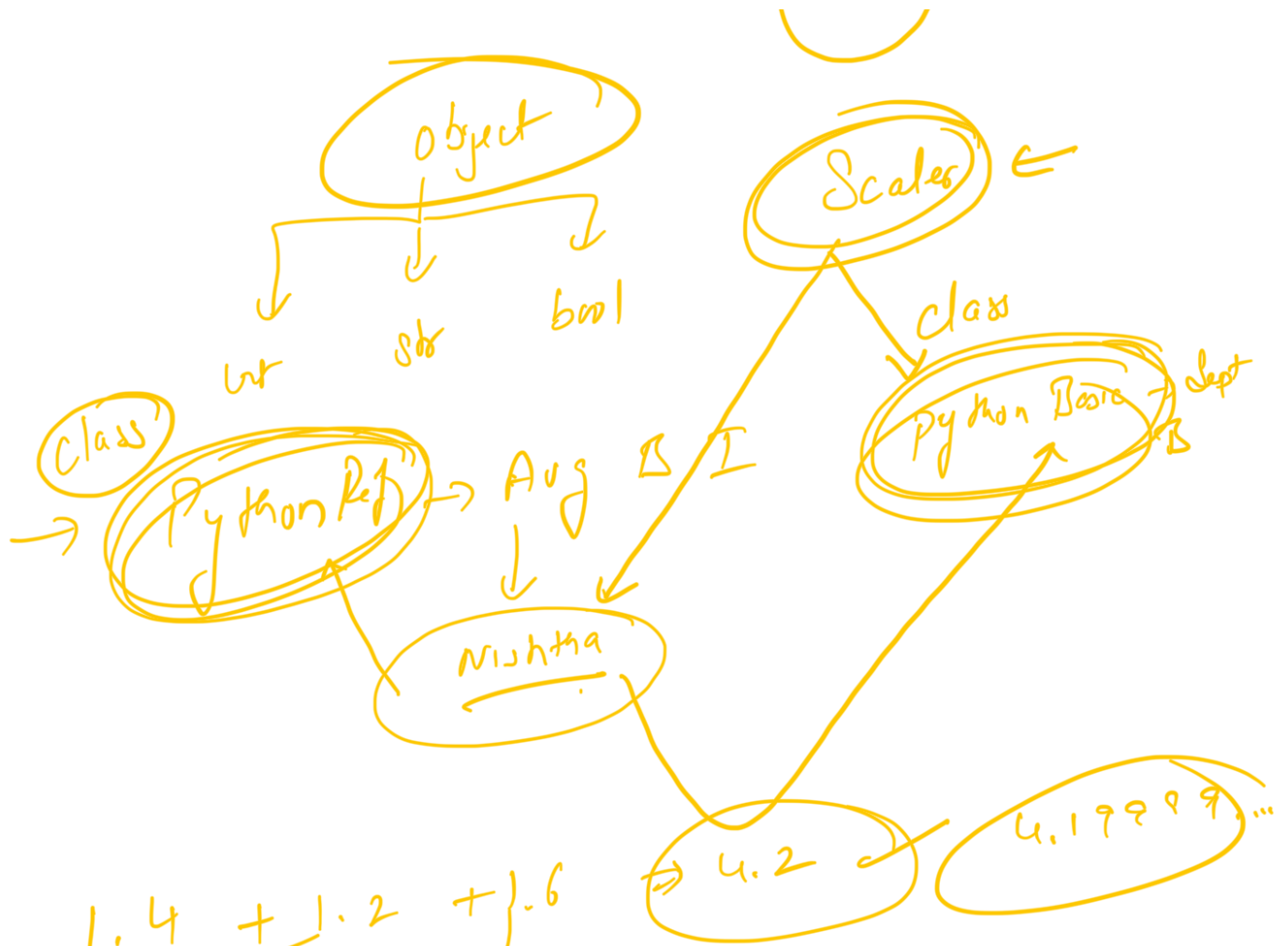


Hierarchy



* Everything in Python that has a value is an object

↳ type(arg) function
 ↳ instance(arg1, arg2)



$$1.4 + 1.2 + 1.6 \Rightarrow 4.2$$

IEEE 754 double precision

$$\begin{array}{r}
 1.3999999911 \\
 1.1999999956 \\
 1.599999 \\
 \hline
 \end{array}$$

$$\begin{array}{r}
 3.333 \\
 3 \overline{) 10}
 \end{array}$$

$$\begin{array}{r}
 9 \\
 10
 \end{array}$$

$$\begin{array}{r}
 9 \\
 10
 \end{array}$$

$$\begin{array}{r}
 2.610 \\
 3 \overline{) 8} \\
 6 \\
 \hline
 20
 \end{array}$$

Floor Division operator

$$\begin{array}{r}
 10 \overline{) 3} \\
 10 \overline{) 3}
 \end{array}$$

$$\begin{array}{r}
 3.3333335 \\
 \hline
 = 3
 \end{array}$$

$$\begin{array}{r}
 -2.66667
 \end{array}$$

$$-8 \mid 3$$



$$2 \times 3 = 6$$

$$2 \times \times 3 = (2)^3 \Rightarrow (2 \times 2 \times 2)$$

$$= 8$$

$$* \quad -a \% b = b - (a \text{ mid } b)$$

$$-14 \% 6 = 6 - (14 \% 6)$$

$$= 6 - 2 = 4$$

$$\Rightarrow (3 \times -2) + 2$$

$$\Rightarrow -6 + 2 \Rightarrow 4$$

$$\Rightarrow (3 \times 2) + 2$$

$$\Rightarrow 6 + 2 \Rightarrow 8$$

$$\begin{array}{r} 3 \overline{) -8} \quad 2 \\ \underline{6} \\ -2 \end{array}$$

$$\begin{array}{r} 3 \overline{) 8} \quad 2 \\ \underline{6} \\ 2 \end{array}$$

$$n \% n \Rightarrow 3 - (8 \% 3)$$



Identifiers

↳ name that we give to our variable.

Rules:-

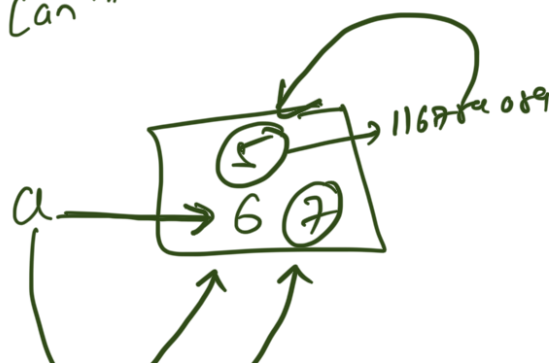
- ↳ it should start with alphabets or underscore (`_`)
- ↳ it can contain alphanumeric or underscore in the middle of name

$abc = 12$ ✓
 $(12)abc = ?$

Mutable | Immutable

↳ whose value can change (list, sets, dictionaries)
 ↳ whose value can't change

$a = 5$
 $n = 6$



Print (a)

a = 5
a = 6 + 1
a = 7

Indented language

2x

a = 10
b = 10

2x

iteration Protocol

↳ Certain datatypes which be iterated upon.

→ iterable : Data structure with atomic elements which can be iterated.
→ iterable : variable used to store atomic elements
→ iteration block : Entire body of loop

a = "Amit"

For (i) in a:
Print(i)

— | —————